

PROJECT REPORT

PROJECT-CALCULATOR

SUBMITTED BY:

NAME: MS AYUSHI

REGISTRATION NUMBER: 26BCE11540

COURSE: CSE CORE (B.TECH)

UNIVERSITY: VIT BHOPAL UNIVERSITY

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1.INTRODUCTION

Project-CALCULATOR is a Python-based command-line calculator developed to perform different mathematical operations. The calculator allows the user to enter numbers and select the required mathematical operation. It can perform basic arithmetic operations as well as several scientific and mathematical calculations.

The project is developed using Python and its built-in math module. The main purpose of this project is to create a simple and user-friendly calculator while applying basic Python programming concepts such as variables, user input, operators, conditional statements, and mathematical functions.

2.OBJECTIVES

- To develop a calculator using Python.
- To perform different mathematical operations.
- To understand user input and conditional statements.
- To use Python's built-in math module.

3. FEATURES

The calculator supports:

- Addition
- Subtraction
- Multiplication
- Division
- Modulus
- Power
- Square root
- Logarithms
- Trigonometric functions
- Factorial
- GCD and LCM
- Rounding and other mathematical operations

4. TECHNOLOGIES USED

- Python 3
- Python Math Module
- VS Code
- GitHub

5. WORKING OF THE PROJECT

The program takes the required numbers from the user and asks for the mathematical operation. It uses conditional statements to identify the selected operation and then displays the result.

The mathmodule is used for advanced mathematical operations such as square root, logarithms, trigonometric functions, factorial, GCD and LCM.

6. TESTING

Operation	Example	Result
Addition	$10 + 5$	15
Subtraction	$10 - 5$	5
Multiplication	10×5	50
Division	$10 \div 5$	2
Square Root	$\sqrt{25}$	5

7. OUTPUT

Enter the first number: 10
Enter the second number: 5
Enter your operation: + 15

8. CONCLUSION

The Project-CALCULATOR successfully performs different mathematical operations using Python. This project helped in understanding Python operators, user input, conditional statements and the mathmodule.